Muhammad Nabeel Malik

Full-Stack AI Engineer | LLM Agents & Full Stack Development

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LinkedIn | GitHub | Portfolio

Professional Summary

Full-stack AI engineer specializing in LLM agents, scalable backends and cloud-based solutions. Building Claire, an AI production agent live for all OSPHIM customers, on Python, FastAPI and Claude. Experienced in delivering industrial SaaS, fintech and e-commerce platforms using React.js, Next.js, Nest.js, PostgreSQL, Kafka, AWS and Docker.

Work Experience

Full-Stack AI Engineer (full-time), *OSPHIM* 07/2026 – Present
Aachen, Germany

- Built Claire, an injection-molding AI agent (Python, FastAPI, SQLAlchemy, Anthropic Claude) on an agent harness with tool calling and tenant-scoped data access; explains jobs, anomalies, downtime and experiment results while flagging data gaps. Live for all customers.
- Built a voice shift-handover pipeline: speech-to-text plus LLM extraction turning spoken handovers into structured, searchable shift data.
- Moved 6 product areas (dashboard, explorer, set points, change log, availability, heat-up) off a 1.4-billion-row, 277 GB raw table onto hourly roll-up and value-history tables: 30-day dashboards 1,150 ms → 16 ms, 90-day explorer filters 2,240 ms → 14 ms, change-log payload 40 MB → 1.4 kB.
- Fixed set-point change detection that only scanned the first 5,000 readings and silently missed changes on long jobs (e.g. found 1 of 4); every change is now detected.

Full-Stack Engineer (part-time), *OSPHIM* 01/2026 – 07/2026
Aachen, Germany

- Rebuilt the production logbook around context: one entry can now link 5+ entity types (machines, molds, jobs, users, production cells), backed by a redesigned PostgreSQL schema and built end to end in Svelte and NestJS.
- Moved all logbook content from plain text to rich JSONB with Drizzle ORM without downtime; my backfill scripts kept 100% of existing tenant entries intact.
- Built a digital shift handover protocol so outgoing shifts hand over open issues and machine status in a structured format instead of paper notes.
- Extended experiment configuration from single-cavity to multi-cavity molds, so process engineers configure and compare all cavities of a mold (e.g. 16) in one setup instead of 16 separate ones.
- Tuned PostgreSQL for 240M+ monthly time-series records in a multi-tenant setup (partitioning, vacuum tuning).

Software Engineer, *MobileLive* 05/2024 – 10/2025
MonieCorp (Fintech Product) Remote

- Developed responsive and modular UI components in React + TypeScript + ShadCN, improving page load speed by 25% and reducing UI bugs by 40% after refactoring.
- Built 15+ SOLID-compliant React components for Admin panel using Cursor IDE for rapid development, cutting client feedback cycles from 3 days to 24 hours.
- Built multi-step React form for admins to configure tiered currency margins with validation; streamlined setup from 20+ minutes to <4 minutes per currency pair.

Staples Canada (MobileLive Client)

- Implemented Mixpanel tracking for 100,000+ users, capturing product views, add-to-cart, and checkout events, enabling business team to optimize flows and boost conversions by 12%.

- Built custom React features using Canva API for Staples Online Print Shop, enabling dynamic product personalization and increasing user engagement by 20%.
- Fixed 30+ production bugs within 4-hour SLAs, maintaining 99.9% uptime during peak sales.

Associate Software Engineer, *Code-Huddle* 08/2022 – 05/2024
GYMYG: Virtual Gym Partner (US-Based Startup) Islamabad, Pakistan
- Built Nest.js APIs with JWT RBAC, handling 10K DAU at <200ms P95 latency.
- Integrated Stripe monthly subscriptions, processing $15k+ in secure payments.
- Reduced video load times by approximately 30% using FFMPEG compression and optimized AWS CloudFront edge delivery.
- Automated end-to-end CI/CD pipeline with Github Actions, AWS CodePipeline, cutting deployment time by 40%.

Ordinami: QR Ordering Web-App (Italian Startup MVP)
- Implemented Firebase Auth + Firestore backend, supporting 100+ daily orders.
- Built an admin-panel shadcn/ui + Tailwind for menu creation, reducing manual setup time for owners by 60%.

Product Development Intern, *Code-Huddle* 04/2022 – 06/2022
Islamabad, Pakistan
- Assisted in frontend (React.js) and backend (NestJS) development, contributing to the design and implementation of scalable solutions.
- Supported AWS operations and collaborated on ongoing projects, gaining exposure to the full product development lifecycle.

Education

Bachelor of Science in Software Engineering, *COMSATS University* 2018 – 2022
Islamabad, Pakistan

Skills and Core Competencies

AI & LLM Engineering
Python | FastAPI | SQLAlchemy | LLM Agents | Agent Harness | Tool Calling | RAG | LLM Extraction | Speech-to-Text

Frontend Development
React.js | Next.js | SvelteKit | Context API | Tailwind CSS | HTML5 | CSS3 | Shopify (Liquid Templating) | ShadCN UI | Redux and Redux Toolkit | RxJs

Software Development & Engineering
Full-Stack Web Applications | RESTful APIs | Microservices Architecture | Secure Authentication Systems | Payment Integrations (Stripe) | CI/CD Automation | SaaS Product Engineering

Cloud, DevOps & Containerization
AWS (ECS, ECR, EC2, RDS, S3, CloudFront, CodePipeline, CloudWatch, SNS) | Azure | Docker

Backend Development
JavaScript | TypeScript | Node.js | Express.js | Nest.js | TypeORM | Prisma | Drizzle ORM | Kafka | Redis

Databases & Data Management
MySQL | PostgreSQL | MongoDB | Query Optimization | Data Modeling

Notable Freelance Projects

Pendfunds.com (M.V.P.), *Contract Management & Mediation Software* 01/2024 – 09/2024
- Developed full escrow payment flow with Stripe Connect and Optimus Bank merchant payouts in Nest.js, onboarding 100+ users and handling hundreds of transactions.
- Automated deployments across development, staging, and production environments using AWS Elastic Beanstalk and AWS CI/CD pipelines.
- Implemented secure payload encryption/decryption using PGP with RSA-256, ensuring sensitive transaction data remained fully protected.